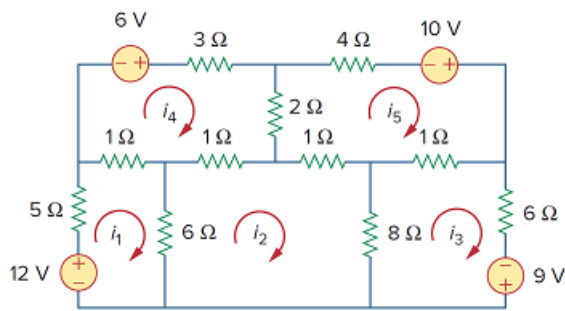


Problem

Use *MATLAB* to solve for the mesh currents in the circuit of Fig. 3.109.

Figure 3.109:



Step-by-step solution

Step 1 of 4

Refer to Figure 3.109 in the textbook.

Apply Kirchhoff's voltage law to the mesh having current i_1 .

$$-12 + 5i_1 + 1(i_1 - i_4) + 6(i_1 - i_2) = 0$$

$$5i_1 + i_1 - i_4 + 6i_1 - 6i_2 = 12$$

$$12i_1 - 6i_2 - i_4 = 12 \quad \dots\dots (1)$$

Apply Kirchhoff's voltage law to the mesh having current i_2 .

$$6(i_2 - i_1) + 1(i_2 - i_4) + 1(i_2 - i_5) + 8(i_2 - i_3) = 0$$

$$6i_2 - 6i_1 + i_2 - i_4 + i_2 - i_5 + 8i_2 - 8i_3 = 0$$

$$-6i_1 + 16i_2 - 8i_3 - i_4 - i_5 = 0 \quad \dots\dots (2)$$

Step 2 of 4

Apply Kirchhoff's voltage law to the mesh having current i_3 .

$$8(i_3 - i_2) + 1(i_3 - i_5) + 6i_3 - 9 = 0$$

$$8i_3 - 8i_2 + i_3 - i_5 + 6i_3 = 9$$

$$-8i_2 + 15i_3 - i_5 = 9 \quad \dots\dots (3)$$

Apply Kirchhoff's voltage law to the mesh having current i_4 .

$$-6 + 3i_4 + 2(i_4 - i_5) + 1(i_4 - i_2) + 1(i_4 - i_1) = 0$$

$$3i_4 + 2i_4 - 2i_5 + i_4 - i_2 + i_4 - i_1 = 6$$

$$-i_1 - i_2 + 7i_4 - 2i_5 = 6 \quad \dots\dots (4)$$

Apply Kirchhoff's voltage law to the mesh having current i_5 .

$$4i_5 - 10 + 1(i_5 - i_3) + 1(i_5 - i_2) + 2(i_5 - i_4) = 0$$

$$4i_5 + i_5 - i_3 + i_5 - i_2 + 2i_5 - 2i_4 = 10$$

$$-i_2 - i_3 - 2i_4 + 8i_5 = 10 \quad \dots\dots (5)$$

Step 3 of 4

Write the equations from (1) to (5) in a matrix form.

$$\begin{bmatrix} 12 & -6 & 0 & -1 & 0 \\ -6 & 16 & -8 & -1 & -1 \\ 0 & -8 & 15 & 0 & -1 \\ -1 & -1 & 0 & 7 & -2 \\ 0 & -1 & -1 & -2 & 8 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \\ i_5 \end{bmatrix} = \begin{bmatrix} 12 \\ 0 \\ 9 \\ 6 \\ 10 \end{bmatrix}$$

Write the MATLAB code to solve the mesh currents.

```
A=[12 -6 0 -1 0; -6 16 -8 -1 -1; 0 -8 15 0 -1; -1 -1 0 7 -2; 0 -1 -1 -2 8];
```

```
B=[12; 0; 9; 6; 10];
```

```
C=inv(A)*B
```

The following is the MATLAB output:

```
C = 2.1701 1.9912 1.8119 2.0942
```

```
2.2489
```

Step 4 of 4

Therefore, from the MATLAB output, the unknown mesh currents are,

$$\begin{array}{l} i_1 = 2.1701 \text{ A} \\ i_2 = 1.9912 \text{ A} \\ i_3 = 1.8119 \text{ A} \\ i_4 = 2.0942 \text{ A} \\ i_5 = 2.2489 \text{ A} \end{array}$$